

Collin Quirke

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SUMMARY

Mechanical Engineering Student with 3+ years of hands-on experience in aerospace hardware design, analysis, manufacturing, and testing. Propulsion Lead Engineer experienced with CAD, FEA/CFD, computational modeling, systems integration, and interdisciplinary development.

EDUCATION

Liberty University, Lynchburg, VA
B.S. Mechanical Engineering (3.3/4.0)

May 2028

EXPERIENCE

Propulsion Lead Engineer, Liberty University Rocketry

July 2026 – Current

- Lead propulsion activities spanning system design, analysis, manufacturing, testing, documentation, and vehicle integration.
- Coordinate propulsion development across aerodynamics, structures, avionics, manufacturing, and vehicle level engineering to manage subsystem interfaces and technical requirements.

Senior Casing Engineer, Liberty University Rocketry

August 2025 – June 2026

- Coordinated propulsion development with aerodynamics, structures, avionics, manufacturing, and vehicle-level engineering to manage subsystem interface requirements.
- Used SolidWorks, ANSYS, MATLAB, and OpenMotor to develop, analyze, and evaluate propulsion system and vehicle hardware.
- Designed and implemented an 8-week structured Solid Rocket Motor onboarding emphasizing hands on development, technical ownership, collaboration, design reviews, and engineering documentation.

Casing Engineer, Liberty University Rocketry

August 2024 – May 2025

- Developed and integrated propulsion-related hardware, including fin-can and boat-tail structures, motor integration, and associated structural interfaces.
- Performed hand calculations and structural analysis to evaluate component loading, factors of safety, and propulsion/vehicle interfaces; used design reviews and post-flight evaluations to inform subsequent iterations.

Research Intern, NASA Virginia Aerospace Science & Technology Scholars (VASTS)

November 2023 – April 2024

- Conducted aerospace research and conceptual mission design through NASA's VASTS program, progressing from individual technical studies to an integrated human Mars mission architecture.
- Developed conceptual designs for long-duration Mars surface experiments, evaluating experimental objectives, resources, hardware, environmental constraints, and operational requirements.
- Investigated technologies and infrastructure for Lunar Gateway operations, incorporating human factors and lunar environmental considerations.
- Collaborated remotely with an interdisciplinary team to develop 6 technical reports, 2 engineering drawings, and orbital mechanic calculations as part of the program's deliverables

PROJECTS

Solid Rocket Motor Design & Analysis Tool – MATLAB

- Developed a tool to automate solid rocket motor conceptual design and analysis, integrating burn-surface geometry, structural calculations, internal ballistics, nozzle analysis, and performance prediction.
- Developed parametrized models for motor casing, propellant grain, and nozzle geometries to rapidly evaluate design configurations and engineering tradeoffs.
- Reduced iteration time by approximately 80% compared with the previous manual workflow.

Level 2 High-Power Rocket – Analysis, Manufacturing & Flight

- Selected and integrated an H-class propulsion system while evaluating structural, aerodynamic, and stability margins.
- Fabricated vehicle components and assembled the propulsive system; post-flight evaluation led to iteration of components.

SKILLS

Design and Simulation: SolidWorks, Onshape, Ansys Mechanical, Ansys Fluent, RocketPropulsionAnalysis, OpenMotor

Languages: C++, Python, MATLAB

Manufacturing: Manual Machining, FDM/SLA 3D Printing, Laser Cutting